NOTRE DAME UNIVERSITY
BANGLADESH

VLSI Design Lab Report

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Submitted to:

Nafisa Tabassum,
Lecturer, Dept of CSE
Notre Dame University Bangladesh

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Abstract

<Write Abstract here>

Software Information

<Write Software information here>

Lab 1 CMOS Logic Gate in DSCH2

1.1 Introduction

<Introduction of Lab-01>

1.2 Objective

<Objective of Lab-01>

1.3 Function

<Write fuction>

1.4 Truth Table

Clock	D	Q(next)
$\uparrow$	0	0
$\uparrow$	1	1
No Edge	X	Q(previous)

1.5 Schematic Image

Figure 1.1: Add figure caption here

1.6 Timing Diagram

Figure 1.2: Add Timing Diagram here

The timing diagram shows that the output Q follows the input D only at the rising edge of the clock signal. Between clock edges, the output remains constant regardless of changes in the input.

From the simulation results, it was observed that the D Flip-Flop changes its output state only at the rising edge of the clock signal. When the clock transitions from LOW to HIGH, the output Q takes the value of the input D. At all other times, the output remains unchanged, preserving the previously stored value. This behavior confirms the correct operation of the edge-triggered D Flip-Flop implemented using CMOS logic.

1.7 Conclusion

<Write Conclusition here>
DEMO
The CMOS-based D Flip-Flop was successfully designed and simulated using the DSCH2 tool. The truth table and timing diagram verified that the circuit operates correctly as a clocked storage element. This experiment reinforces the understanding of sequential logic design and the role of flip-flops in digital systems.

Lab 2 Implementing Sequential Circuit

<Lab 2 Start here>

Lab 3 Adder Circuits

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Lab 4 Flip-Flop Circuits

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Lab 5 Stick Diagram in MICROWIND2

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Lab 6   Stick Diagram of XOR function

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